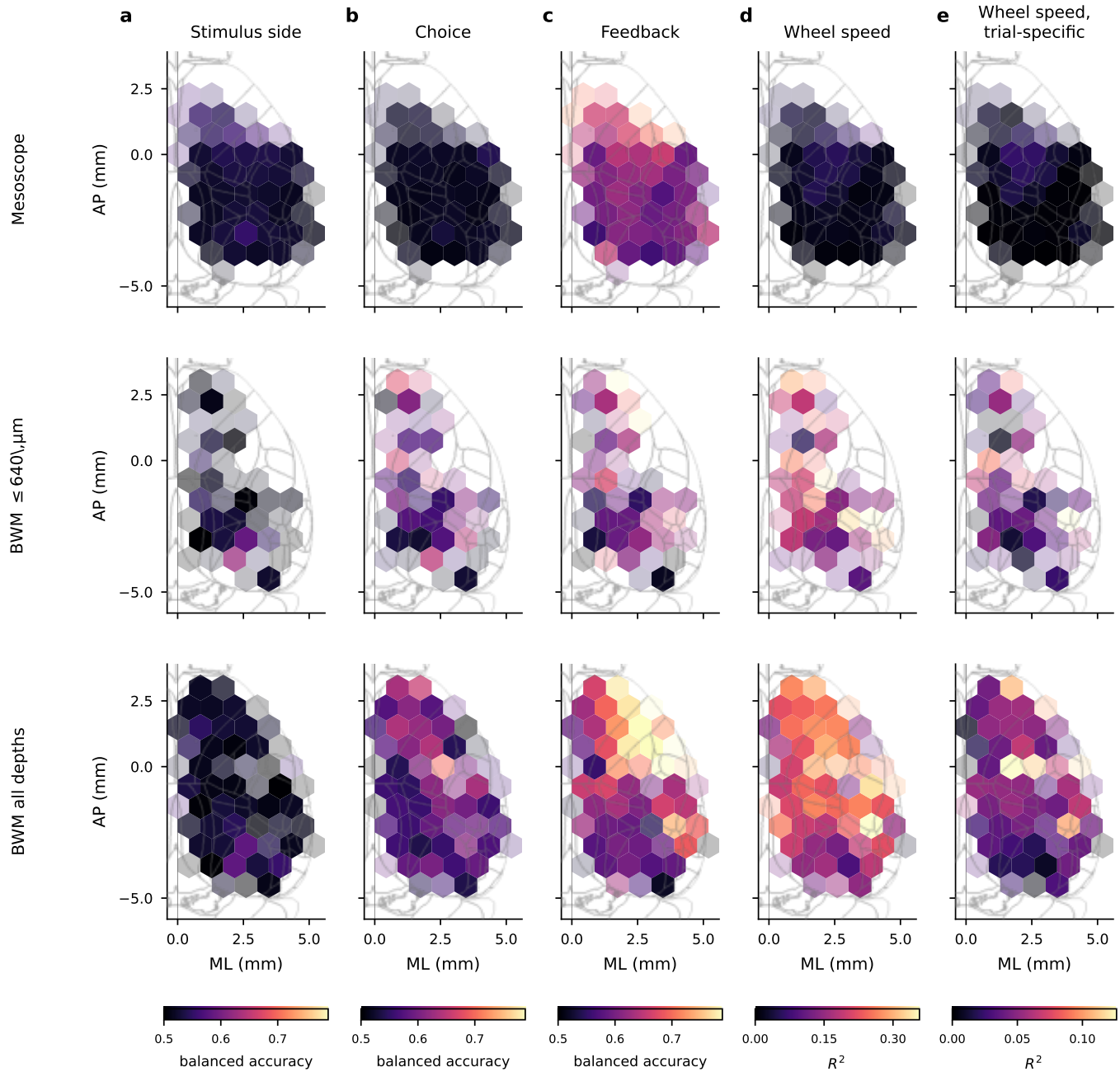


Figure 1: Dorsal cortex maps (revised): mesoscope two-photon imaging (top row; deconvolved activity of isocortex neurons) vs Brain Wide Map Neuropixels units within 640 μm of the pial surface (middle row; twice the depth range the mesoscope images, 174–318 μm) and at all depths (bottom row; reference). Each neuron is assigned to a hexagonal bin (1 mm vertex to vertex) by its ML and AP Allen CCF position; hemispheres folded onto the right. Grey lines: Allen top-view region boundaries. Colour scales are shared across rows. Decoding per session and bin (sessions with $\geq$50 trials and $\geq$10 neurons in the bin), 5-fold cross-validation over trials. Binary variables: L2 logistic regression, balanced accuracy (chance 0.5). Wheel speed in six 0.2 s bins from -0.2 to 1 s around movement onset: ridge regression on the activity of the previous, same and next bin (penalty by exact leave-one-trial-out error), cross-validated R^2; trial-specific: the same with the mean speed profile of the training trials subtracted, so only trial-to-trial differences in speed count. 10 randomly drawn neurons per session and bin (mean of 3 draws), so rows are compared at the same population size. Opacity: number of animals contributing to the bin (full opacity at $\geq$4 animals; fainter bins rest on fewer animals). **a** Stimulus side decoded from 10 randomly drawn neurons, averaged across sessions; window $[0, 0.1]$ s (BWM) or $[0, 0.3]$ s (mesoscope) relative to stimulus onset. **b** Choice; window $[-0.1, 0]$ s (BWM) or $[-0.3, 0]$ s (mesoscope) relative to movement onset. **c** Feedback; window $[0, 0.2]$ s (BWM) or $[0, 0.3]$ s (mesoscope) relative to feedback. **d** Wheel speed, cross-validated R^2. **e** Trial-specific wheel speed (stereotyped speed profile after movement onset removed), R^2.